

ALX Professional Foundations (PF): Week #6 Milestone Worksheet

SECTION A: User Interview Preparation

Step 1: Restate your team's problem statement

Please write your team's agreed-upon problem statement from Week 6 here:

Step 1: My Team's Problem Statement

In the Sahel region of Sub-Saharan Africa, nomadic pastoralist communities, particularly women and children, face severe challenges in accessing essential maternal and child healthcare services due to their mobile lifestyle, remoteness, and cultural barriers. This lack of access results in high rates of maternal and infant mortality, preventable diseases, and malnutrition, perpetuating a cycle of poverty and vulnerability.

Without accessible and reliable healthcare services, women and children within these communities struggle to receive necessary medical attention, leading to preventable health issues and exacerbating existing inequalities.

The problem disproportionately affects women and children in nomadic pastoralist communities, who face unique obstacles due to their transient lifestyle, limited healthcare infrastructure, and cultural practices that may hinder access to medical care.

Addressing this issue requires the development of a comprehensive mobile health platform that connects these communities with essential maternal and child healthcare services. A successful solution will involve mobile health clinics, culturally sensitive healthcare providers, and community engagement to build trust and address barriers to access, ensuring equitable healthcare delivery and improving health outcomes for women and children in the Sahel region.

Step 2: Define the User Persona

Define the person you're solving the problem for in as specific terms as possible. This is important as this person is the one who will ultimately benefit from your solutions, and you'll be

interviewing someone who fits these criteria. This means identifying the characteristics, objectives, motives, and pain points of your target users. In the space below, define this user by creating a detailed user persona of them using the following descriptions:

- **Demographics:** Age, gender, location, education, family status, interests, hobbies and more
- **Pain points:** What the user might have trouble with, like being not very tech savvy. Identify at least 2.
- **Goals:** What the user wants to achieve using your product or what are their goals with respect to the problem they're facing.
- **Motivations:** What motivates the user to use a product or solve their problem. How motivated are they to receive a potential solution?
- **Behavior:** How the user might behave in the context of the problem or when a solution is presented to them.
- **Customer needs and wants:** What the customer might need or want as a potential solution to their problem.

Step 2: User Persona

Demographics:

- **Age:** 18 years old
- **Gender:** Female
- **Location:** Nomadic pastoralist community near Dori, Sahel region, Burkina Faso
- **Education:** Limited formal education, primarily focused on herding and traditional knowledge. Attended primary school sporadically due to her family's migratory patterns.
- **Family Status:** Recently married, expecting her first child
- **Interests:** Learning about maternal health, ensuring a healthy pregnancy and safe delivery, supporting her family's livelihood, preserving cultural traditions
- **Hobbies:** Assisting with herding livestock, storytelling, traditional crafts
- **Occupation:** Stay-at-home mum who does a bit of seasonal farming due to their nomadic lifestyle & traditional crafts

Pain Points:

- **Limited Access to Healthcare:** Isabel's community is constantly on the move, making it difficult to access healthcare facilities. The nearest clinic is a day's journey away, and transportation is often unreliable and expensive.
- **Cultural Barriers:** Isabel's family and community hold traditional beliefs about pregnancy and childbirth, which may discourage her from seeking modern medical care. She may also face stigma and discrimination due to her young age and pregnancy.

- **Fear and Anxiety:** Isabel is worried about the health of her unborn child and her own well-being during childbirth. She lacks access to reliable information and support, leading to fear and anxiety.

Goals:

- Have a healthy pregnancy and safe delivery.
- Give birth to a healthy baby.
- Learn about maternal and child health to provide the best care for her child.
- Become a respected and knowledgeable mother in her community.

Motivations:

- Strong desire to be a good mother and protect her child.
- Fear of complications during pregnancy and childbirth.
- Desire to learn and improve her knowledge of maternal and child health.
- Sense of responsibility towards her family and community.

Behavior:

- Respectful of elders and traditional practices, but also open to new information and ideas.
- Resourceful and adaptable, accustomed to a challenging environment.
- Seeks advice and support from her family and community.
- Eager to learn and utilize new technologies that can benefit her and her family.

Customer Needs and Wants:

- Accessible and culturally appropriate maternal and child healthcare services, ideally brought directly to her community through mobile clinics.
- Healthcare providers who speak her language (Fulfulde) and understand her cultural context.
- Reliable information and education on pregnancy, childbirth, newborn care, and family planning, delivered through channels like mobile apps or community workshops.
- A supportive network of women who can share experiences and provide emotional support.
- Empowerment to make informed decisions about her health and the health of her child.

Further issues to think on:

- May have limited literacy skills, so any technology or information provided should be visually appealing and easy to understand.
- She may be hesitant to use technology if she doesn't see it as relevant or beneficial to her life.
- Building trust and rapport with her community is crucial for any intervention to be successful.

Step 3: Interview Questions

Now that you have your user persona defined, let's prepare for the interview by defining the questions that you'll be asking during the interview. To do so, provide the list of questions that intend to ask during the interview in the space below. These questions should help you understand the problem and how it affects the target users better. Write down at least 5 primary questions that you're going to ask.

Step 3: Interview Questions

Section 1: General Information

1. Could you please tell me your age?
2. What is your primary occupation?
3. How many children do you have?

Section 2: Healthcare Experiences

1. Can you describe your experience accessing healthcare during your pregnancies and after giving birth?
2. What were the main challenges you faced in getting the healthcare you needed during those times? (Examples: distance, transportation, cost, cultural beliefs, language barriers)
3. How does your mobile lifestyle impact your ability to access healthcare for yourself and your children?
4. What are the main health concerns you have for yourself and your children?

Section 3: Healthcare Preferences and Needs

1. What kind of healthcare services would you like to have better access to? (Examples: antenatal care, postnatal care, vaccinations, family planning)
2. What kind of information or support do you feel is most important for your health and the health of your children?

3. How do you feel about traditional healers and practices compared to modern healthcare? When would you choose one over the other?
4. What are your expectations of healthcare providers? How can they better meet your needs?
5. If you could design a healthcare system that worked best for you and your community, what would it look like?

Section 4: Technology and Healthcare

1. Do you have access to a mobile phone? If so, how often do you use it?
2. In what ways do you think technology could be used to improve your access to healthcare? (Examples: mobile clinics, telehealth, informational apps)

Section 5: Feedback and Recommendations

1. What advice would you give to organizations like ours that are trying to improve healthcare for nomadic communities?
2. Is there anything else you'd like to share about your experiences or concerns regarding healthcare?

SECTION B: User Interview Insights

Step 1: Interviewee Information

Please write the name and other details of the interviewee you interviewed for the information.

Step 1: Interviewee Information

Interviewee Name: **Isabel Ouedraogo**

Interviewee Occupation: Housewife

Interviewee Age & Location: 18 , Nomadic pastoralist community near Dori, Sahel region, Burkina Faso

Step 2: Interview Insights

What did you learn from the interview? Provide the main points that you gathered through the interview about your problem. On the whole, what was new that you learnt about your problem

and its effect on people? What were the most pressing concerns and pain points mentioned in the interviews? What recommendations did you receive from the interviewees about possible solutions? Did you uncover any new factors surrounding the problem that you didn't consider before? All in all, identify at least 5 major themes with details.

Step 2: Interview Insights

Section 2

1. Healthcare Access Experiences:

- **Distance and Transportation:** All respondents noted significant challenges due to the long distances to healthcare facilities. Transportation was often unreliable or unaffordable, especially in emergencies.
- **Cost:** Highlighted the high cost of healthcare services, which made accessing even primary prenatal and postnatal care difficult.
- **Language Barriers:** Half of the respondents faced difficulties communicating with healthcare providers due to language differences, leading to misunderstandings and inadequate care.
- **Cultural Beliefs:** Several women mentioned that traditional beliefs and practices sometimes delay or discourage seeking professional medical help.

2. Main Challenges:

- **Distance and Transportation:** Consistently mentioned as a primary obstacle, the distance to healthcare facilities and lack of reliable transportation were significant barriers.
- **Cost:** The financial burden of healthcare was a significant issue, with many unable to afford necessary services.
- **Cultural Beliefs:** Traditional beliefs often influence healthcare decisions, creating a barrier to accessing timely medical care.
- **Language Barriers:** Communication issues due to language differences were also a notable challenge for many women.

3. Impact of Mobile Lifestyle:

- **Irregular Access:** The nomadic lifestyle of many respondents resulted in inconsistent access to healthcare facilities, affecting the continuity of care for both mothers and children.
- **Missed Appointments:** Frequent movement led to missed prenatal and postnatal appointments, impacting the health of both mothers and infants.
- **Vaccinations:** Ensuring children receive timely vaccinations was a common concern due to the mobile nature of their lives.

4. Main Health Concerns:

- **For Mothers:**

- Complications during pregnancy and childbirth were significant worries due to limited access to consistent prenatal care.
- Post-birth infections and complications were also mentioned as primary concerns.
- **For Children:**
 - Malnutrition and preventable diseases, such as malaria, were major worries for mothers.
 - Ensuring timely vaccinations and regular health check-ups for children was a significant concern, exacerbated by their mobile lifestyle.

Section 3

1. Desired Healthcare Services:

- **Antenatal Care:** Respondents emphasized the need for regular check-ups during pregnancy.
- **Postnatal Care:** Follow-up care after childbirth was mentioned as crucial for both mother and baby's health.
- **Vaccinations:** Ensuring children receive timely vaccinations is a priority for all.
- **Family Planning:** Many respondents highlighted the importance of having access to family planning services and information.

2. Important Information and Support:

- **Nutrition Advice:** Guidance on proper nutrition for pregnant women and children was a common need.
- **Health Education:** Information on recognizing symptoms of common illnesses and when to seek medical help was frequently mentioned.
- **Breastfeeding Support:** Support and advice on breastfeeding were essential to many respondents.
- **Clean Water and Sanitation:** Information on maintaining hygiene to prevent infections and diseases was considered essential.

3. Views on Traditional Healers vs. Modern Healthcare:

- **Cultural Significance:** Traditional healers are essential in the community and are trusted for specific ailments and cultural practices.
- **Preference for Serious Conditions:** All respondents preferred modern healthcare due to its proven effectiveness for more severe health issues, especially during pregnancy and for children.
- **Choosing Traditional Healers:** Traditional healers were often selected for minor ailments and cultural practices, while modern healthcare was sought for critical and complex health issues.

4. Expectations of Healthcare Providers:

- **Respect and Compassion:** Respondents expected healthcare providers to treat them with dignity and understand their cultural contexts.

- **Clear Communication:** The need for healthcare providers to explain health issues and treatments in a language they understand was a common expectation.
- **Accessibility:** Respondents emphasized the importance of healthcare services being geographically and financially accessible.
- **Supportive Care:** Continuous care and follow-up services, especially post-childbirth, were essential expectations.

5. Ideal Healthcare System:

- **Local Clinics:** Accessible clinics in each community to reduce travel distance was a unanimous suggestion.
- **Mobile Health Units:** Regular visits from mobile units to serve nomadic and remote populations were considered essential.
- **Affordable Services:** Subsidized healthcare costs to ensure affordability for all community members.
- **Cultural Sensitivity:** Training healthcare providers to understand and respect cultural beliefs and practices.
- **Comprehensive Care:** A range of services, including antenatal and postnatal care, vaccinations, family planning, and health education, were highlighted as necessary.
- **Community Health Workers:** Local health workers provide primary care, education, and support, bridging the gap between the community and healthcare facilities.

Section 4: Technology and Healthcare

1. Access to Mobile Phones:

- **Mobile Phone Ownership:** All respondents reported having access to a mobile phone.
- **Frequency of Use:** Those with mobile phones use them daily for communication, primarily to connect with family members and occasionally to access information.

2. Technology to Improve Healthcare Access:

- **Mobile Clinics:** All respondents saw the potential benefits of mobile clinics, which could bring healthcare services directly to their communities, reducing the need to travel long distances.
- **Telehealth Services:** Many respondents were open to telehealth consultations, where they could receive medical advice and consultations via phone or video, especially for follow-up appointments and minor health concerns.
- **Informational Apps:** Respondents expressed interest in informational apps that could provide health education, reminders for vaccinations and check-ups, and tips on maternal and child health. These apps could be handy in areas with limited access to healthcare facilities.

Section 5: Feedback and Recommendations

1. Advice for Organizations:

- **Cultural Sensitivity:** Understand and respect nomadic communities' cultural practices and beliefs. Engage with community leaders and traditional healers to gain their support and trust.
- **Accessibility:** Focus on bringing healthcare services closer to the communities through mobile clinics and regular health camps. Ensure services are affordable or subsidized.
- **Education and Awareness:** Provide health education and awareness programs tailored to the community's needs and languages. Use visual aids and local dialects to communicate effectively.
- **Community Involvement:** Involve community members in planning and implementing healthcare initiatives to ensure they meet the actual needs and preferences of the community.

2. Additional Experiences and Concerns:

- **Continuity of Care:** Respondents expressed concerns about the continuity of care due to their nomadic lifestyle. They suggested maintaining health records that can be accessed from different locations.
- **Trust in Healthcare Providers:** Building trust with healthcare providers was highlighted as crucial. Consistent and respectful interactions with providers can help build this trust.
- **Emergency Services:** The need for reliable emergency healthcare services was a recurring concern. Respondents recommended establishing a system for quick response to medical emergencies, possibly through mobile health units equipped for emergencies.

SECTION C: Generating Solutions

Step 1: Meeting Date, Time, & Location

Please list when and where your team meeting took place.

Step 1: Meeting Date, Time, & Location

- A. Date: 21-07-2024
- B. Time: 17:00 GMT +2
- C. Location: Online (Zoom)

Step 2: Meeting Attendees

Please list who attended your team meeting, and their primary role.

Step 2: Meeting Attendees

1. Zinzile Mhlanga - Project Manager
2. Tapiwa Setimela - Data Analyst
3. Boemo Phirinyane - Data Analyst (primary)
4. Christina Tawfiq - UX Researcher
5. Michelle Ngarandi - Product Manager

Step 3: Bad Idea Brainstorm

It's time to start thinking about solutions to the problem. Use all the information you now have about the problem (from your research last week and the interviews this week) to start thinking of possible solutions. As you have studied in Canvas modules, it's always good to first gather as many ideas as possible. So at this stage, don't hold back, put your divergent thinking hat on, and let the creativity flow to gather as many ideas as possible. As a team, you must generate at least 10 new bad ideas. Remember, the dumber the idea, the better! This is to help you work as a team to be non-critical. Stay in divergent thinking. It helps to say "thank you" after every idea is shared.

Step 3: Brainstormed Ideas

- Camel-Drawn Mobile Clinics: Outfit camels with medical supplies and send them on their own to reach remote villages.
- Baby Delivery Drones: Develop drones equipped with medical kits to airdrop midwives and supplies during childbirth emergencies.
- Telepathy Health Consultations: Train healthcare workers in telepathy to conduct remote consultations with patients without needing phones or internet access.
- Singing Vaccination Reminders: Create catchy jingles about vaccinations and broadcast them across the Sahel using loudspeakers mounted on trucks.
- Herbal Medicine Vending Machines: Install solar-powered vending machines dispensing traditional herbal remedies in remote villages.
- Mommy and me app /website : Expectant mother's can use to make doctor appointments, receive tips e.g diet, exercise, ask their healthcare provider questions when needed ,alerts on healthcare events in the community , etc

- Medical-Ride App/usd - A platform where mothers can book for a ride to the clinic based on their appointments or emergencies.
- Mobile health clinics: equipped vehicles that travel with nomadic communities, providing essential healthcare services, vaccinations, prenatal and postnatal care, and nutritional support.
- Health Information Systems: Systems for collecting and managing health data across nomadic populations, aiding in resource allocation.
- Telemedicine Platforms: Digital platforms enabling remote consultations with healthcare providers via mobile devices.
- Training Programs: Digital platforms for training the individuals in the community enabling them to provide basic healthcare and education.
- Digital Health Records: Implement electronic health record systems that are accessible across various healthcare providers and mobile clinics. This ensures continuity of care, even as nomadic families move across different regions.
- Satellite Communication: Utilise satellite technology for reliable internet connectivity in remote areas, enabling telemedicine consultations, access to healthcare databases, and real-time communication between healthcare providers and nomadic communities.
- Community Engagement Platforms: Develop digital platforms e.g mobile messaging groups via sms that facilitate community engagement, health education, and feedback mechanisms to improve healthcare delivery. This can be used by local leaders and the information passed to the health providers

Step 4: Team's Final Selected Solution Idea

Your next task is to narrow your choices, which will put you in a convergent thinking mindset. You should have some discussion and debate about this, and try to reach a consensus on a final solution to your problem that your team is going to consider working on for the rest of Month 2. These ideas can be totally new, or they can be the same or variations from ideas you've already come up with. Remember that they should involve some sort of technology (either a piece of software like an app or algorithm, or a physical device such as a robotic fish or machine that scans your DNA). You will not have to build the solution out. But you will have to create some type of basic prototype (if it is a device) or a set of wireframes (if it is an app/software). You will not have to actually create the technology or code.

You must figure out a fair way to reach a consensus with your group, including a discussion where everyone's voice can be heard.

Step 4: Team's Final Selected Solution Idea

Website / App Name: Sahel Mamatoto (Mother and Child)

Solution: A comprehensive mobile health platform that connects nomadic pastoralist

communities in the Sahel with essential maternal and child healthcare services.

SECTION D: Product Planning

Step 1: Product Description

You learnt about product planning and product descriptions in Weeks 4 and 5. Now it's time to apply that learning to create these descriptions to plan for your solution. In the space below, describe the solutions that you're building, in as much detail as possible. Ask yourself the following questions:

- What does the ideal solution look like? Will it be an app or a physical item or a software service? What will it look like aesthetically?
- How the ideal solution will function, and how will users interact with it? Will the users create profiles? Will there be a dashboard (and what will it show)? Will there be other forms of screens or interactions that users will perform? How will users operate the product?
- What will be the features of the solution? How will you define and describe these features and how will users access these features on the app or physical product?

Step 1: Product Description

Product Description: Sahel Mamatoto (Mother and Child) – A Digital Lifeline for Nomadic Communities

Sahel Mamatoto is a comprehensive mobile health platform designed to empower nomadic pastoralist women and families in the Sahel region by providing accessible, culturally relevant, and reliable maternal and child healthcare services. It aims to bridge the gap between remote communities and essential healthcare through a combination of innovative technology, community engagement, and culturally sensitive care.

The ideal solution would be a mobile application (app) due to its accessibility, scalability, and ability to provide real-time information and communication. Aesthetically, the app should be:

- **User-friendly:** Simple and intuitive interface suitable for users with varying levels of digital literacy.
- **Culturally sensitive:** Incorporate visuals and language that resonate with nomadic pastoralist communities, respecting their cultural beliefs and practices.

- **Accessible:** Designed to function well on lower-end smartphones and in areas with limited internet connectivity.

Functionality and User Interaction:

1. **User Profiles:** Users (mothers) will create profiles upon initial setup, including basic demographic information, pregnancy status, and health history.
2. **Dashboard:** The app will feature a personalized dashboard showing:
 - Appointment reminders for prenatal check-ups.
 - Notifications for important health alerts or updates relevant to pregnancy and postnatal care.
 - Access to educational resources on maternal health, nutrition, and child care.
3. **Appointment Booking:** Users can schedule appointments with healthcare providers directly through the app, considering the availability and proximity of providers.
4. **Teleconsultations:** Facilitate virtual consultations with healthcare providers for remote diagnosis, advice, or follow-up care.
5. **Emergency Support:** Provide quick access to emergency contacts or services in case of urgent medical needs.
6. **Health Records:** Enable mothers to maintain digital health records including prenatal check-ups, vaccinations for children, and postnatal care details.
7. **Language Support:** Offer multilingual support to bridge communication gaps between users and healthcare providers.

Features of the Solution:

- **Appointment Scheduling:** Allow users to book appointments with nearby healthcare providers based on their location.
- **Health Education:** Provide access to articles, videos, and interactive content on maternal and child health, tailored to cultural sensitivities.
- **GPS Integration:** Use GPS to locate nearby healthcare facilities and provide directions or transportation options.
- **Offline Access:** Ensure core functionalities like emergency contacts and basic health information are accessible even without internet connectivity.
- **Feedback Mechanism:** Allow users to provide feedback on healthcare services received, helping improve service quality.

Operation of the Product:

Users will operate the app primarily through their smartphones:

- **Downloading and Setup:** Users will download the app from app stores or through community health worker outreach programs.

- **Registration:** Upon installation, users will create profiles and input relevant information.
- **Daily Use:** Users will interact with the app to access health information, schedule appointments, receive notifications, and manage their health records.

Additional Features (Potential for Future Development):

- **Offline Mode:** Enable access to certain features and information offline, considering the limited internet connectivity in some areas.

Step 2: Product Solution

Before we finalize everything for the week, it's also important to very clearly define how your product is going to solve the problem that you set out to solve. You can do so by answering the following questions:

- What specifics about the product or app contribute to solving the problem?
- How do these specific features contribute to solving the problem?
- How does the product help the people you're creating the solution for?

Step 2: Product Solution

Product Solution

Sahel Mamatoto: A Digital Bridge to Maternal and Child Health

How Sahel Mamatoto Addresses the Problem:

Sahel Mamatoto tackles the core issues faced by nomadic pastoralist communities in accessing maternal and child healthcare through a multi-pronged approach:

1. Overcoming Distance and Accessibility Barriers:

- Mobile Health Clinics: Sahel Mamatoto partners with local healthcare providers to deploy mobile clinics equipped with essential medical supplies and personnel. These clinics travel to remote communities, providing on-the-ground care and eliminating the need for long and dangerous journeys.

- Telemedicine Consultations: The app's telemedicine feature allows women to connect with healthcare providers remotely, bypassing the need for travel and enabling timely consultations for both urgent and routine care.

2. Addressing Cultural and Linguistic Barriers:

- Culturally Sensitive Content: The app's health education library is tailored to the specific cultural context and beliefs of nomadic communities, ensuring that information is relevant, understandable, and respectful of local traditions.
- Translation Services: Telemedicine consultations are available in local languages, ensuring clear communication and building trust between healthcare providers and patients.
- Community Health Workers: The app empowers local women as community health workers, who serve as trusted liaisons between the community and healthcare providers. They can offer culturally sensitive education and support, helping to bridge the gap between traditional practices and modern medicine.

3. Empowering Women with Information and Support:

- Health Education Library: The app provides a wealth of information on maternal and child health, nutrition, family planning, and disease prevention. This empowers women to make informed decisions about their health and the health of their children.
- Community Forum: The platform fosters a sense of community and support, allowing women to share experiences, ask questions, and learn from each other. This can reduce isolation and stigma, and promote a culture of collective well-being.
- Vaccination Reminders and Tracking: The app's reminder system ensures that children receive timely vaccinations, a critical factor in preventing childhood diseases.

4. Leveraging Technology for Improved Outcomes:

- Geolocation and Mapping: By tracking the movement patterns of nomadic communities, healthcare providers can anticipate healthcare needs and proactively deploy resources, ensuring that care is available when and where it is needed most.
- Emergency Alerts and Response: The app's emergency alert feature enables quick and efficient responses to medical emergencies, potentially saving lives.
- Data Collection and Analysis: The platform's data collection capabilities allow for continuous monitoring and evaluation of health outcomes and service utilization, facilitating data-driven decision-making and program improvement.

How Sahel Mamatoto Helps Its Users:

- Improved Access to Healthcare: Provides convenient and reliable access to essential maternal and child healthcare services, regardless of location.
- Culturally Sensitive Care: Delivers healthcare services that respect and integrate traditional beliefs and practices.

- Empowerment through Information: Equips women with the knowledge and resources to make informed decisions about their health and the health of their children.
- Community Support: Fosters a sense of community and belonging, providing a platform for women to connect and support each other.
- Enhanced Health Outcomes: Reduces maternal and child mortality rates, improves nutrition, and prevents the spread of disease.
- Economic Empowerment: By improving health and well-being, the app contributes to greater economic productivity and participation for women and families.

Step 3: Reflections (Individual)

Please share your **personal** reflections on your experience with your team so far.

Step 3: Team Process Reflections

A. What is working well with your team?

Teamwork and collaboration

B. What is one good thing that happened during your team meeting?

We had good understanding and agreed on all the points discussed

C. What is one thing your team could do better in the next meeting?

To have the meetings earlier in the week

D. Are you experiencing any concerns or frustrations with your team? If yes, what can you personally do to lessen the concern/frustration?

No

E. How would you rate your ability to communicate with your team members on a scale of 1 to 4? (1=extremely poor and 4=excellent)

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F. Overall, how satisfied are you with how well your team is working together? (On a scale of 1 to 4, with 1=extremely poor and 4=excellent)

G. 4

H. Is there anything else you'd like to share about your team and their process?

.thank you for giving me the opportunity to work with this wonderful team